COMPUTER ENGINEERING

QUEZON CITY

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| **COURSE CODE/TITLE:** | CPE 010 | **SECTION:** | CPE21S2 |
| **DATE PERFORMED:** | 09/02/2026 | **DATE SUBMITTED:** | 09/02/2026 |
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**ACTIVITY NO.** **Prelim Examination**

## 1. PROCEDURE OUTPUT

| Conditions | Best-Case Time Complexity | Worst-Case Time Complexity | Brief explanation |
|---|---|---|---|
| 1. Finding an Element | O(1) | O(N) | For loop creates N number of iterations while if the target is in the first index, iteration only happens once |
| 2. Printing All Elements | O(N) | O(N) | Displaying all items iterates all the elements of an array which means, only O(N) occurs making it both the best and worst case time complexity. |
| 3. Nested Loops | $O(N^2)$ | $O(N^2)$ | When a nested loop occurs, an iteration of the outer loop automatically iterates the inner loop. Meaning $O(N^2)$ is both the worst and best case time complexity. |
| 4. Conditional Processing | O(1) | O(N) | If the target is less than 100, the execution only happens once, making it the best time complexity. If the target is greater than 100, N number of iterations happen. |
| 5. Fixed Inner Loop | O(N) | O(N) | The inner loop is independent of the outer loop since it only iterates 10 times each iteration of i. In the given code there are separate conditions to trigger the iterations of the inner and outer loop which is why the best and worst time complexity is the same. |
| 6. Two Consecutive Loops | O(N) | O(N) | Both loops run simultaneously which is why O(2N) occurs. This is basically the same as O(N). It is both the worst and best time complexity. |
| 7. Dividing by Two | O(log N) | O(log N) | If initial conditions aren't met, then O(1) occurs which is the best case time complexity. However, since there are no early exits. If initial conditions |

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| | | | are met, $O(\log_2 N)$ will occur since it will continue to execute until the value of x is less than or equal to 1. |
| 8. Nested Loop With Different Limits | O(N) | $O(N^2)$ | If i = 0 or all negative numbers, then O(N) will occur since 0 or all negative numbers will not be less than 0, The iteration only runs on the I array and not on the J array. Otherwise, $O(N^2)$ occurs if both conditions are met such as if i is a non-zero positive integer. |
| 9. Early Termination | O(1) | O(N) | If the target is found in the first index, the loop executes only once. If the target is not found in the first index, it will loop for N times until the target is found. |
| 10. Two-Level Search | O(N) | $O(N^2)$ | If the target is at the front, then it only loops for N times since the second loop will break.. However, if the target is at the Nth index, then the loop will continue to run in an N*N pattern until it finds the target. |
| 11. Linear Loop With Logarithmic Inner Work | O(N log N) | O(N log N) | The for loop runs for N times and consistently satisfies the conditions for the $\log_2 N$ operation. Which is why it works like a nested loop such as N * $\log_2 N$ operations is satisfied. |
| 12. Conditional Nested Loop | O(1) | $O(N^2)$ | If the target is found at the front index, it will only loop once. However if that item is at the Nth position it may trigger a condition for a nested loop or N*N operations to happen. |
| 13. Finding the Maximum | O(N) | O(N) | Since finding the maximum of the array needs the system to loop the whole list at least once, both the worst and best complexity is O(N). |
| 14. Three Nested Loops | $O(N^3)$ | $O(N^3)$ | In a three way nested loop, if i iterates, then j iterates, then k also iterates making a single execution run times. Therefore N*N*N is $N^3$. |
| 15. Comprehensive Analysis | O(log N) | O(log N) | If at least one iteration is done, then both of the conditions for the for loop and while loop is met, then log of N occurs. |

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:** **Gemini**

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** **Explanation of concepts**

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:** **I generated the idea first then ask AI if my interpretation was correct.**

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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